

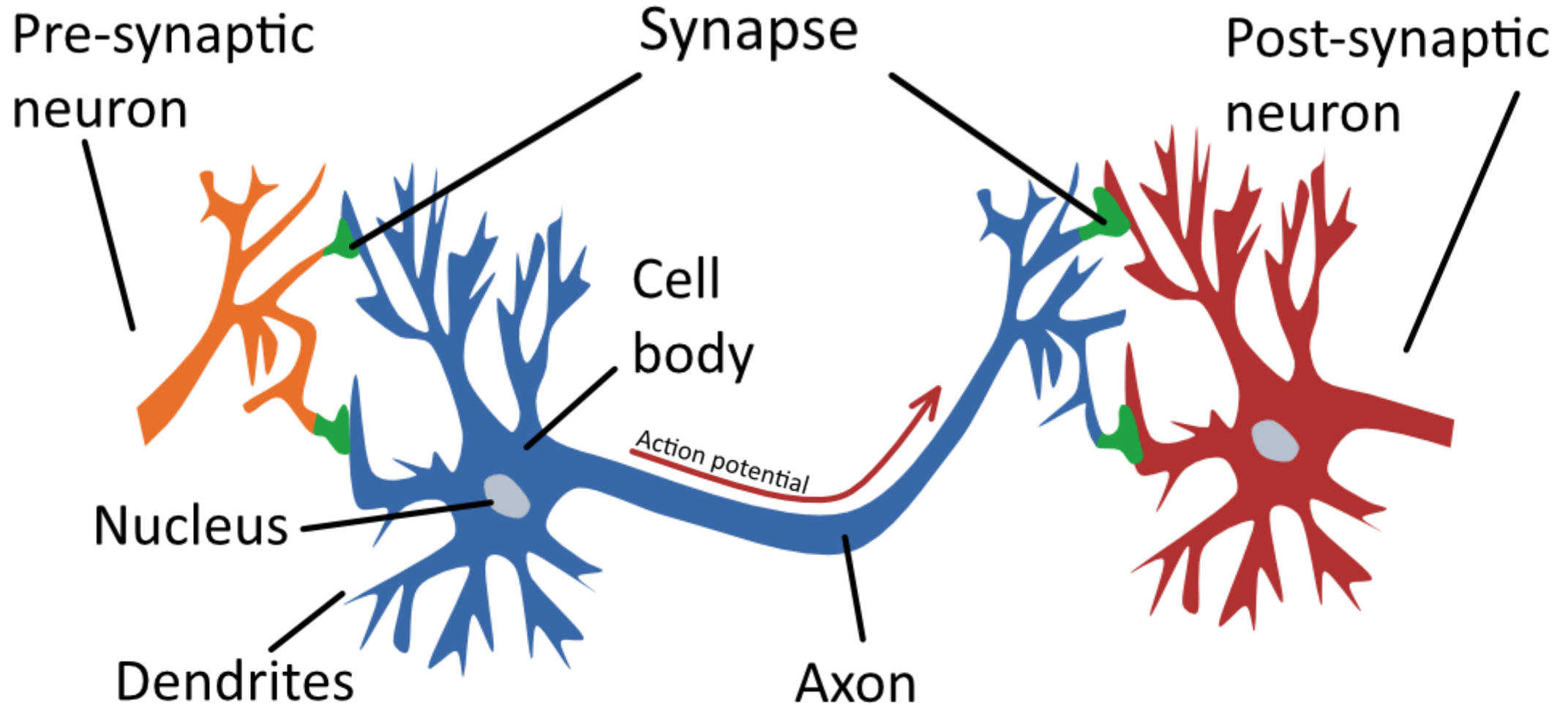
Lecture week 2: Neural network basics

Iris Proff

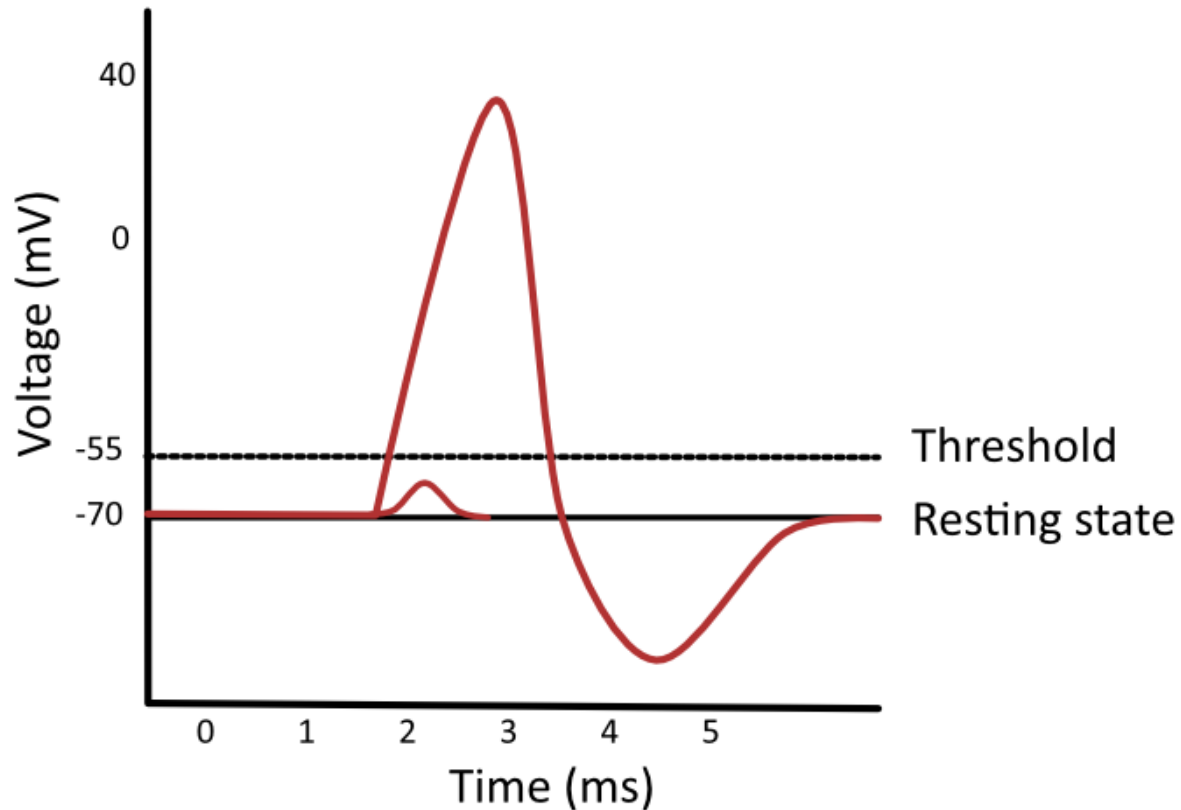
Video 1: Artificial neurons

Video 2: Hopfield networks

A biological neuron – structure

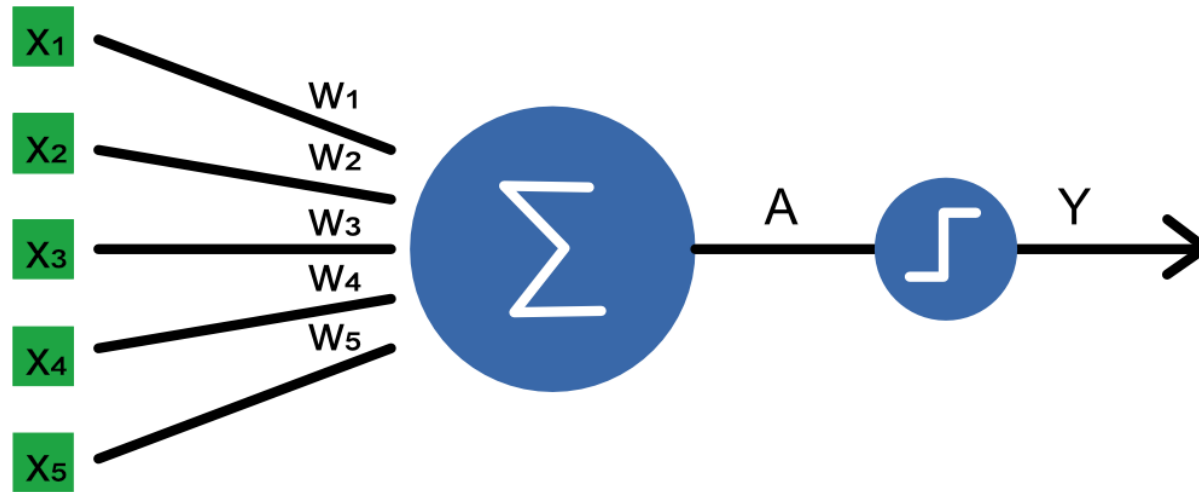


The action potential



1. Through input from pre-synaptic neurons, voltage in cell body starts to rise
2. If the voltage exceeds a threshold of approx. -55mV, neuron spikes – action potential
3. This electrical signal is passed along the axon to post-synaptic neurons
3. After an action potential, the neuron is hyperpolarized – refractory period
4. Neuron returns to resting state

The artificial McCulloch-Pitts Neuron

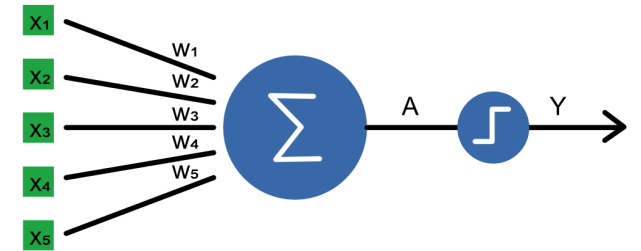


1. Inputs x_{1-5} are multiplied by synaptic **weights** w_{1-5}
 $w_j > 0$ excitatory synapse
 $w_j < 0$ inhibitory synapse
2. Neuron sums the weighted inputs to **activation A**
3. Nonlinear activation function is applied to arrive at **output Y**

McCulloch-Pitts - equation

We can compute the output of the neuron as follows:

$$Y = \text{sgn}\left(\sum_{i=1}^n w_i x_i - b\right),$$



where b is the activation threshold or bias of the neuron.

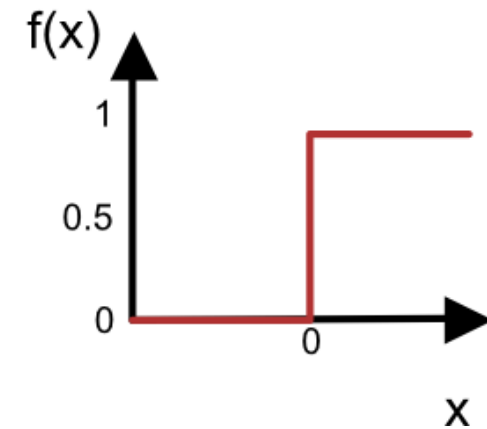
The sign function (“sng”) transforms the neuron’s activation into 1 or 0:

$$Y = 1, \text{ if } \sum_{i=1}^n w_i x_i > b \quad \text{and} \quad Y = 0, \text{ if } \sum_{i=1}^n w_i x_i < b$$

Types of activation functions

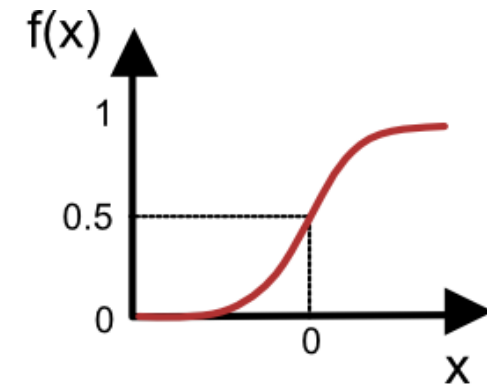
Threshold / sign function

$$f(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$



Sigmoid function

$$f(x) = \frac{1}{1 + e^{-x}}$$



Advantage: sigmoid function is differentiable

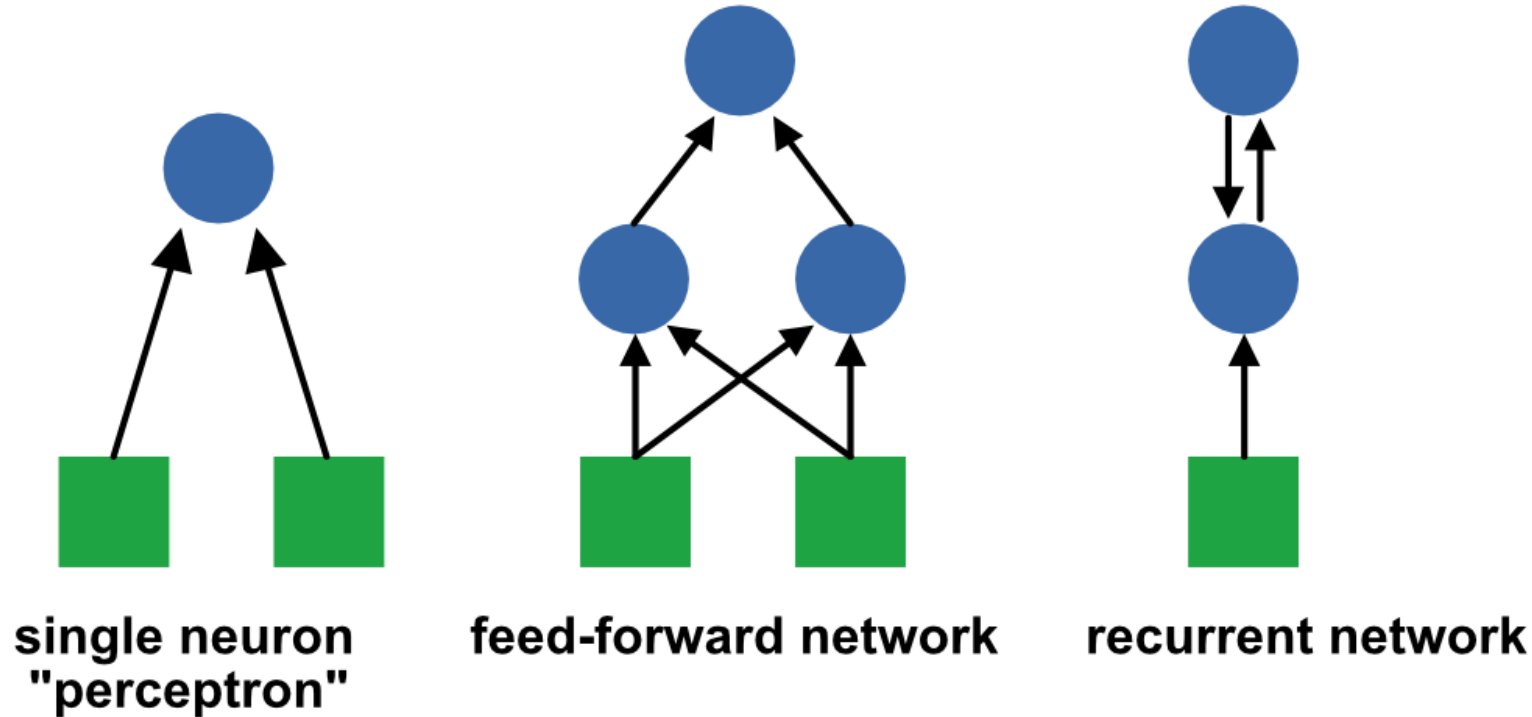
But: the neuron now has a gradual response

→ interpretet as the neurons average activity

→ „rate based“

Networks of artificial neurons

One neuron cannot do much of its own. But we can wire multiple McCulloch-Pitts neurons to different types of neural networks:



Question for you:

Which features of biological neurons are reflected in the McCulloch-Pitts Neuron?

What are simplifications of the model?

Biological neuron	McCulloch-Pitts Neuron
Receives inputs through synapses with different strength	Receives inputs through weighted connections
...	...

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Video 1: Artificial neurons

Video 2: Hopfield networks

Auto-associative memory

Some amazing properties of memory in the brain

- Auto-associative / content addressable: remember someone's name by the smell of their perfume, remember song by first notes
- Capable of handling noisy and incomplete information
“pattern completion”
- Robust against loss of cells, “graceful degradation”
- High learning capacity and flexibility



How can we build a model of that?

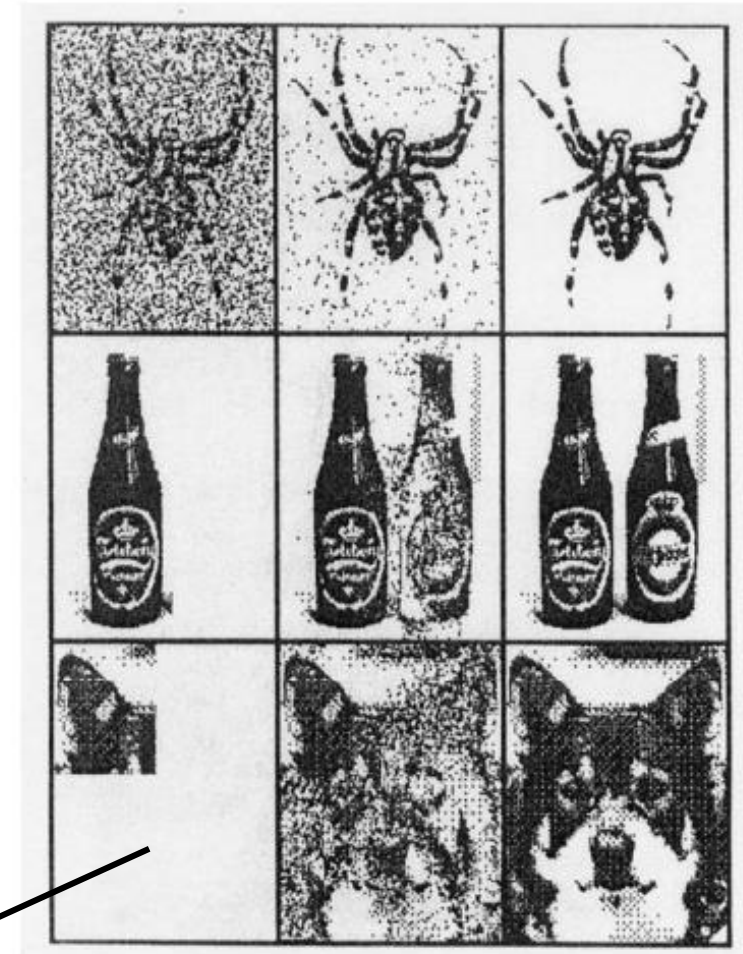
Hopfield network

John Hopfield (1982) considered McCulloch-Pitts Neurons in a recurrent network architecture

Hopfield networks are able to store and recall patterns of activation

→ Neural model of memory

Every pixel corresponds to one neuron which is either on (black) or off (white)



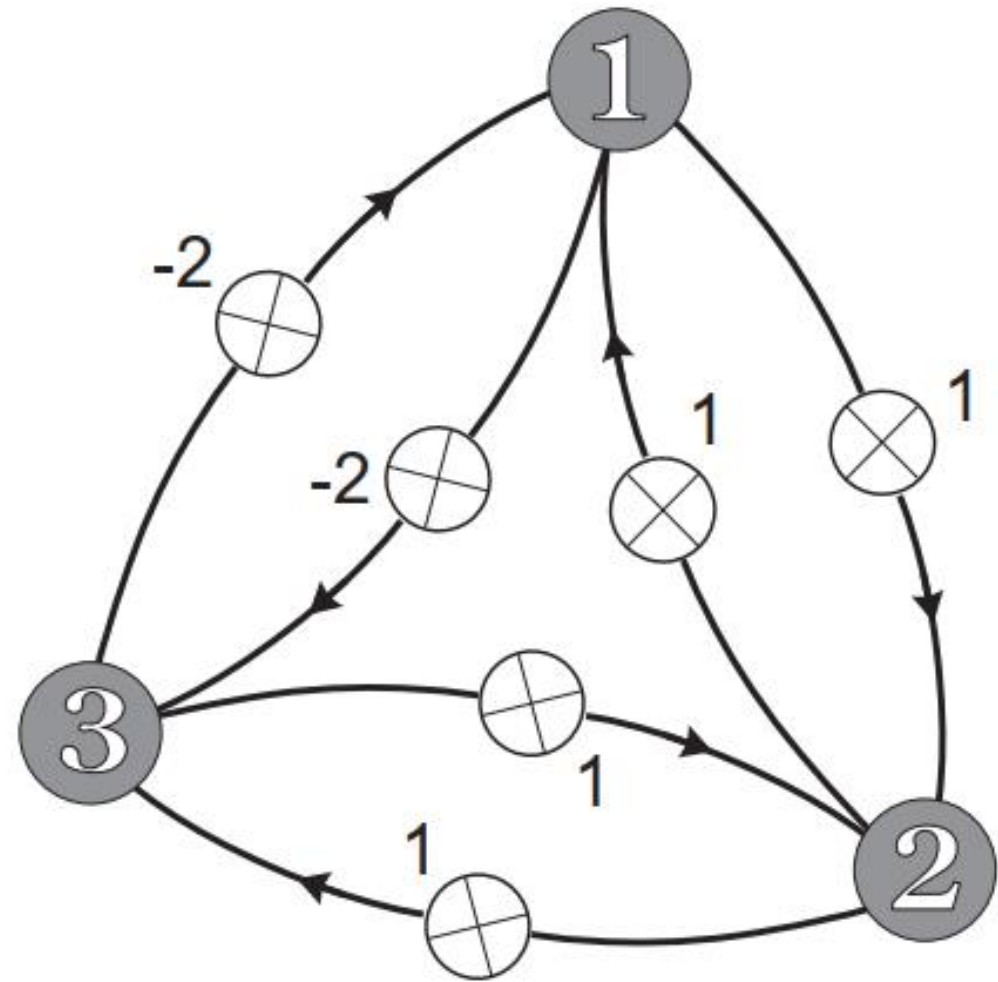
Hertz, Krogh & Palmer, 1991

Hopfield network

- Outputs are $(-1, 1)$
- Threshold $b = 0$
- Step function as activation function
- Every neuron is connected to every other neuron (but not itself) and weights are symmetric:

$$w_{ij} = w_{ji}$$

- Asynchronous updating: at every step pick a random neuron and update its activation



A Hopfield network with 3 neurons

Value of a Hopfield Neuron

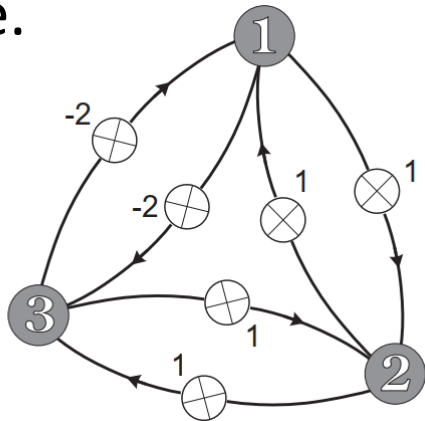
$$u_i = \sum_{j \neq i} w_{ji} v_j, \quad v_i = \begin{cases} 1, & u_i \geq 0 \\ 0, & u_i < 0 \end{cases}$$

u is the activation of the neuron, v is the value of the neuron

State transition table

A network of 3 neurons can be in $2^3 = 8$ states.

At each time point, one of the three neurons is updated, yielding a new network state.

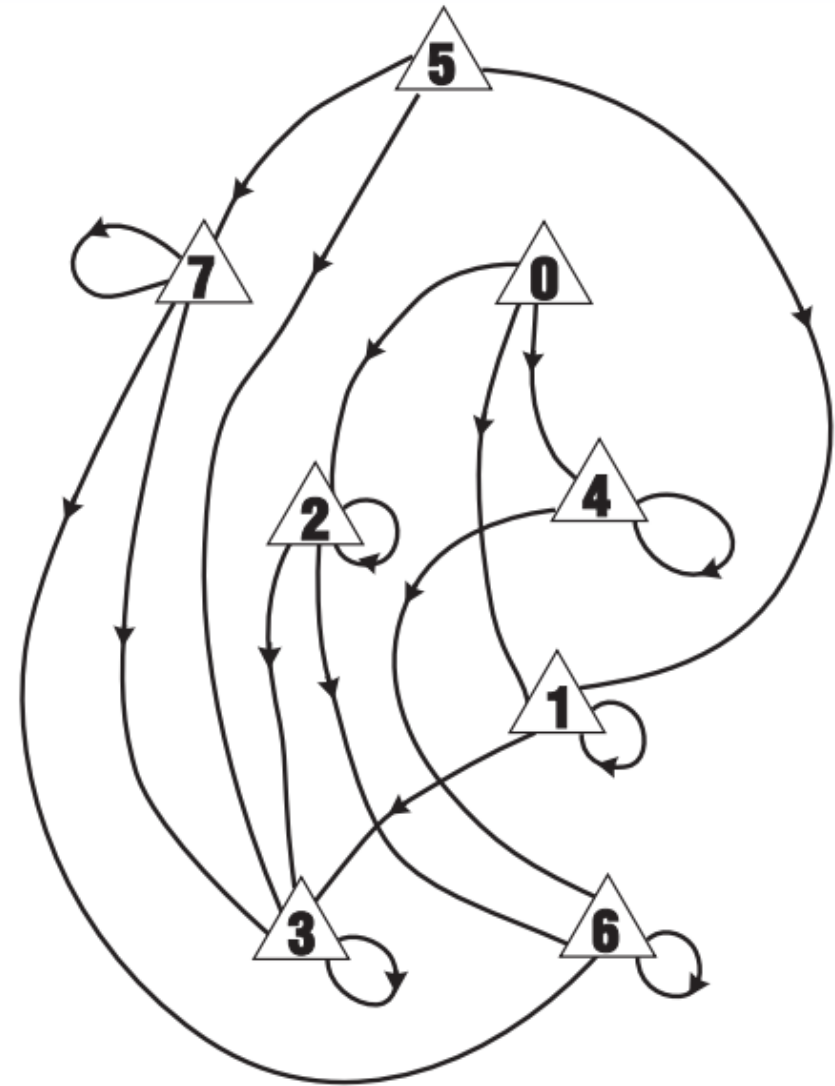


Old state of network				New state after updating neuron		
State number	Neurons					
	x_1	x_2	x_3	x_1	x_2	x_3
0	-1	-1	-1	4	2	1
1	-1	-1	1	1	3	1
2	-1	1	-1	6	2	3
3	-1	1	1	3	3	3
4	1	-1	-1	4	6	4
5	1	-1	1	1	7	3
6	1	1	-1	6	6	6
7	1	1	1	3	7	6

State transition diagram

The same information can also be summarized in a state transition diagram.

It shows which state transitions are possible by updating one of the neurons.



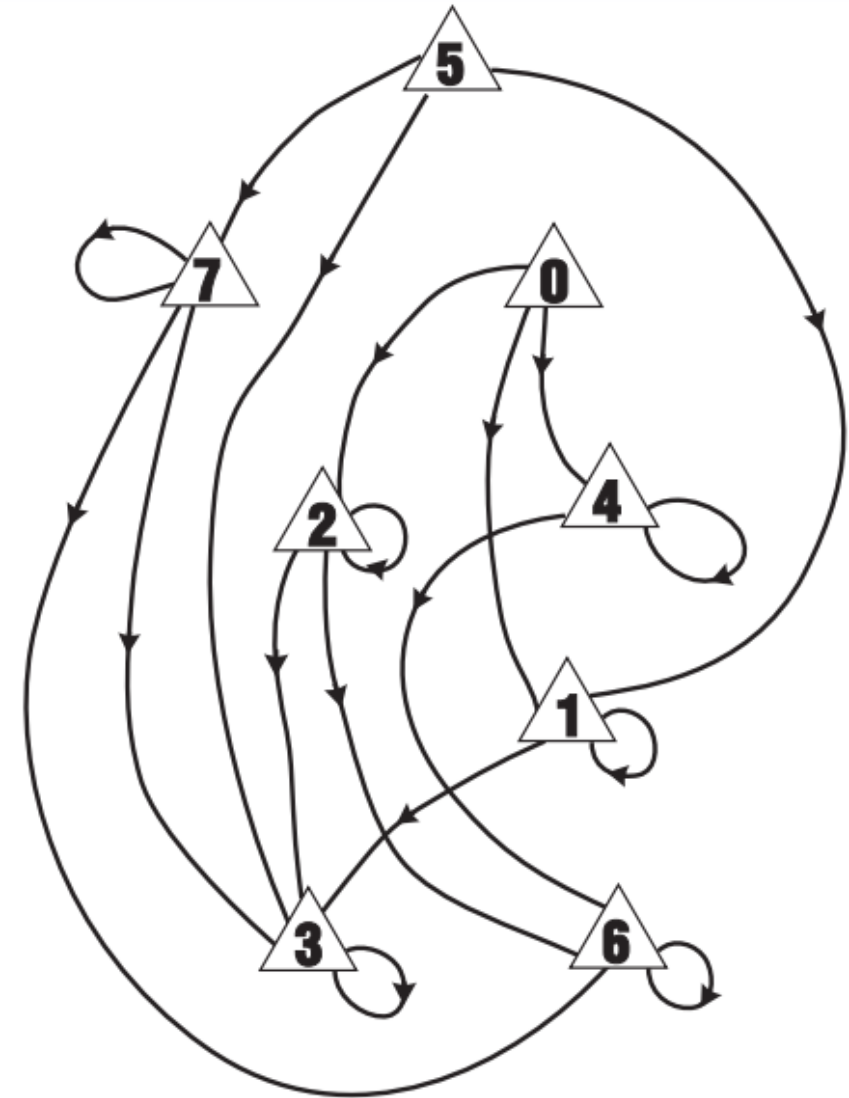
Attractor states

- The interesting thing about Hopfield networks is that their states don't just keep changing.
- Given an initial pattern, the Hopfield network will always settle on a fixed state after a number of iterations.
- We call this an **attractor state**.

Attractor states

Can you spot the attractor state(s)?

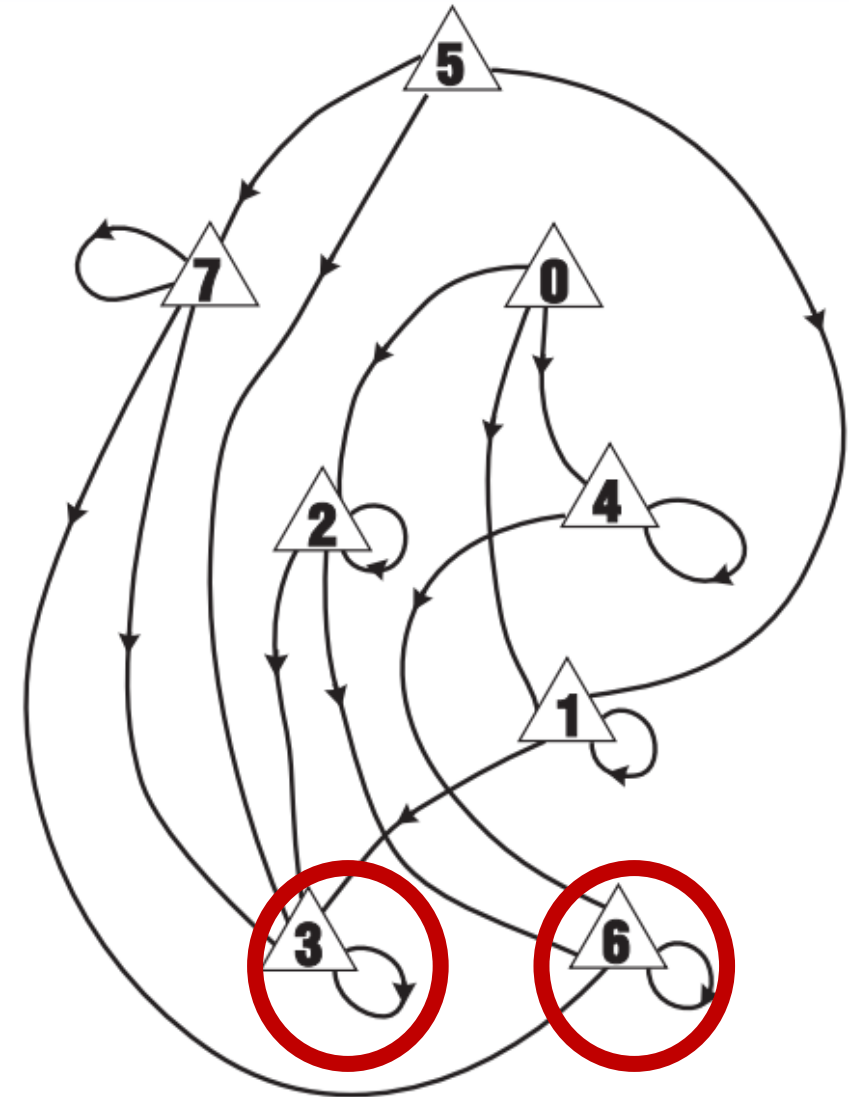
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Attractor states

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Consistency

As the network approaches an attractor state, the output of the neurons becomes more consistent with the weights.

The consistency of a Hopfield net can be measured as:

$$\sum_i \sum_j w_{ij} v_i v_j$$

Consistency – 2 simple examples

Consider these 2 networks made of 2 connected neurons

If $v_i = v_j = 1$,

$$\text{consistency} = 1 * 1 * 1 = 1$$

If $v_i = v_j = -1$,

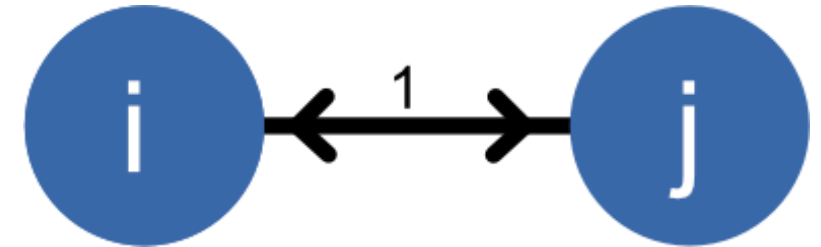
$$\text{consistency} = 1 * -1 * -1 = 1$$

If $v_i = -1, v_j = 1$,

$$\text{consistency} = 1 * 1 * -1 = -1$$

If $v_i = 1, v_j = -1$,

$$\text{consistency} = 1 * -1 * 1 = -1$$



If $v_i = v_j = 1$,

$$\text{consistency} = -1 * 1 * 1 = -1$$

If $v_i = v_j = -1$,

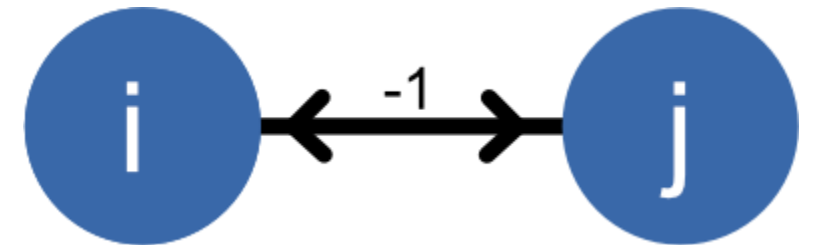
$$\text{consistency} = -1 * -1 * -1 = -1$$

If $v_i = -1, v_j = 1$,

$$\text{consistency} = -1 * 1 * -1 = 1$$

If $v_i = 1, v_j = -1$,

$$\text{consistency} = -1 * -1 * 1 = 1$$



“Flip-Flop”

Consistency – 2 simple examples

Consider these 2 networks made of 2 connected neurons

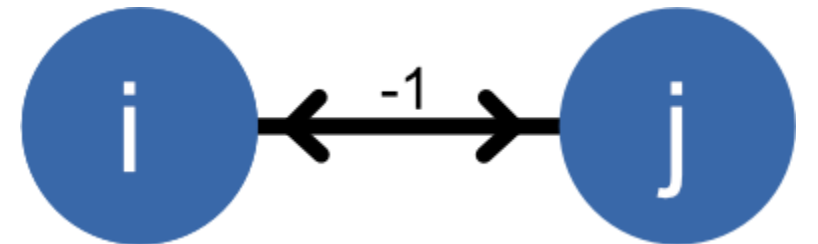
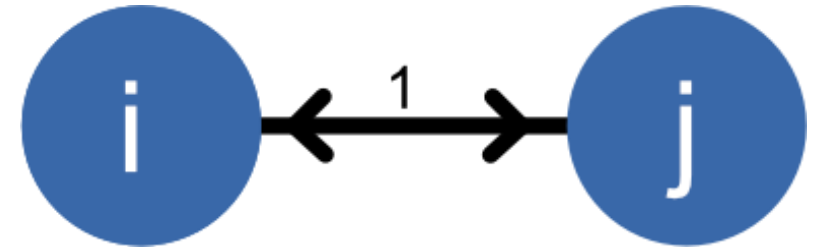
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Attractor states

If $v_i = v_j = 1$,
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consistency = $1 * 1 * 1 = 1$
consistency = $1 * -1 * -1 = 1$
consistency = $1 * 1 * -1 = -1$
consistency = $1 * -1 * 1 = -1$

consistency = $-1 * 1 * 1 = -1$
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“Flip-Flop”

Energy function

A network in an attractor has low energy.

A network in transition has high energy.

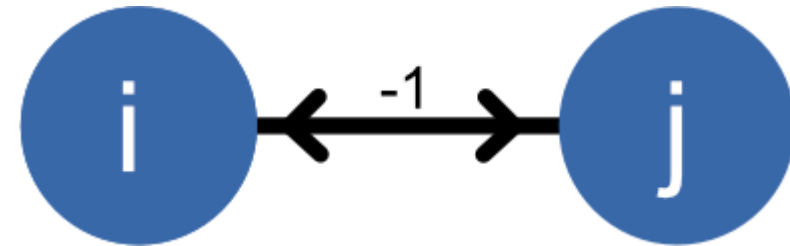
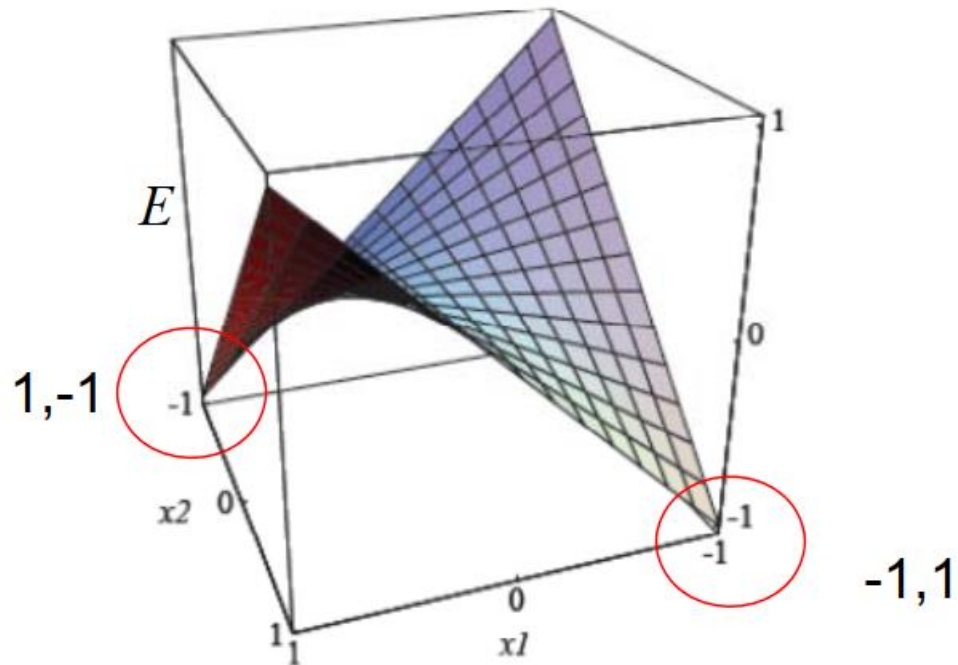
→ opposite of consistency

$$E = -\frac{1}{2} \sum_i \sum_j w_{ij} v_i v_j$$

As the Hopfield network approaches the attractor state, it minimises the energy function.

Flip Flop energy function

$$E = -\frac{1}{2} * (-v_1 v_2 - v_1 v_2) = v_1 v_2$$

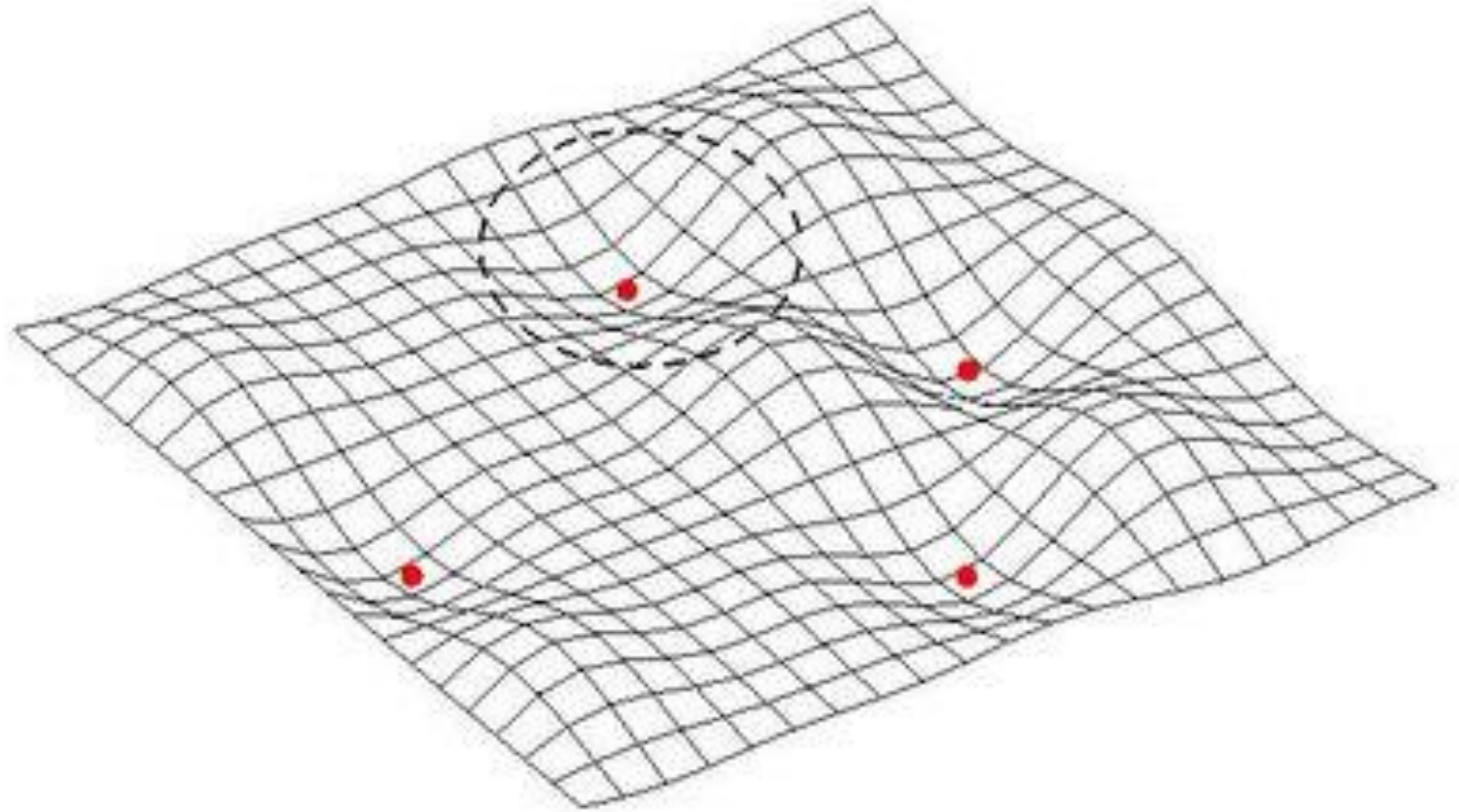


Pattern recall

Attractor states are local minima (“valleys”) in the energy landscape.

A Hopfield network can have multiple attractor states.

It will converge to / recall the one that is closest to the input pattern.



Learning in a Hopfield Network

But how can we store patterns (“memories”) and make them attractors?

Storing memories the brain



- **Hebb's rule:** „what fires together wires together“ – synaptic plasticity
- Synaptic connections between neurons that are activated together get stronger
→ **long-term potentiation (LTP)**
- When a neuron is stimulated sub-threshold for an extended period through a synapse, the synaptic connection is weakened
→ **long-term depression (LTD)**

Learning in a Hopfield Network

- Learning rule to store pattern p : $w_{ij} \leftarrow w_{ij} + p_i p_j$
- Strength of connection between two active neurons strengthened
→ long-term potentiation / Hebb's rule
- Strength of connection between active and inactive neuron weakened
→ long-term depression
- When all patterns have been stored, divide each weight by the number of patterns to normalize the weights.

Emergence

Emergence = an entity shows properties that its parts do not have on their own, but that result from complex interaction of the parts

Emergent properties of Hopfield networks

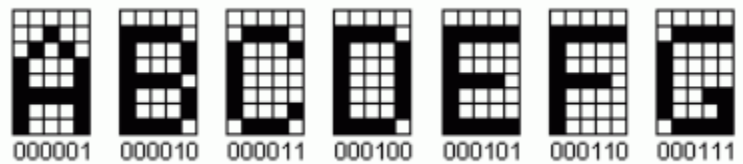
- Stable attractor states
- Pattern completion
- Familiarity recognition
- Noise tolerance
- Graceful degradation

The bridge between simple circuits and the complex computational properties of higher nervous systems may be the **spontaneous emergence** of new computational capabilities from the collective behavior of large numbers of simple processing elements.

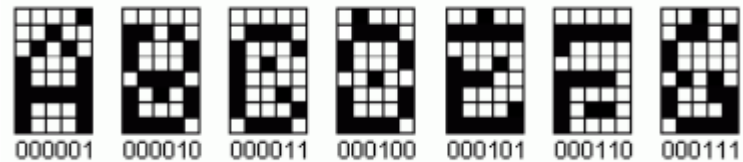
Hopfield, 1982

Pattern recall - example

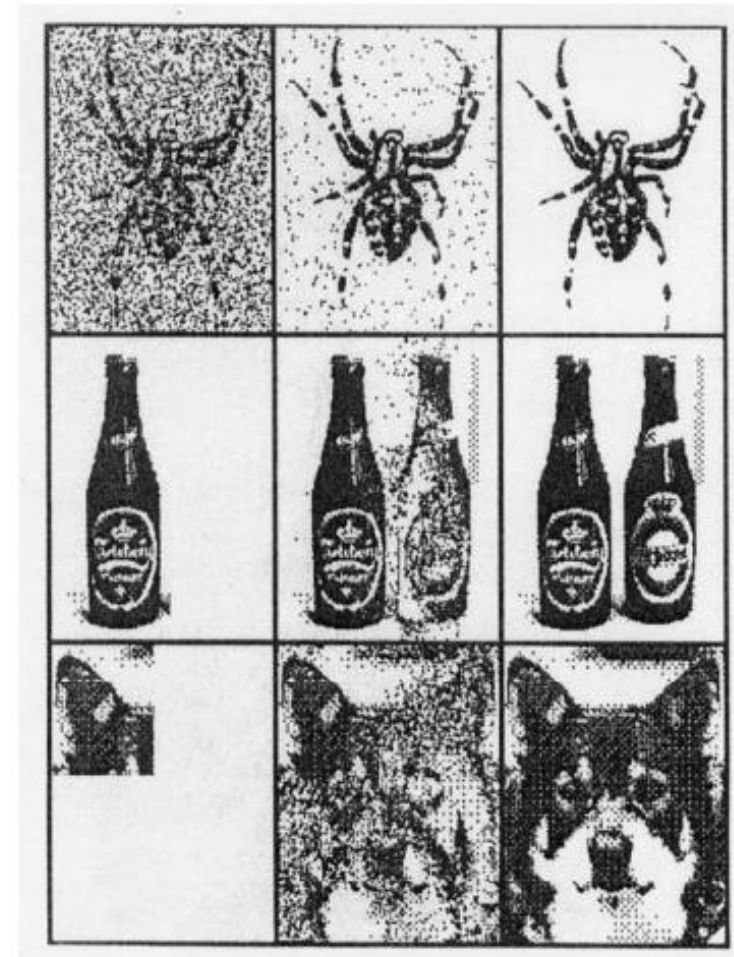
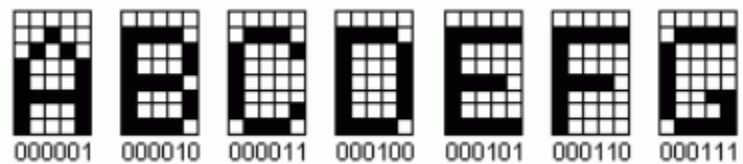
Training patterns



Present corrupted patterns



Network reconstructs original patterns



Hertz, Krogh & Palmer, 1991

Reading for this week

How do attractor networks relate to memory formation in the Hippocampus?

Rennó-Costa, C., Lisman, J. E., & Verschure, P. F. (2014). A signature of attractor dynamics in the CA3 region of the hippocampus. PLoS Comput Biol, 10(5), e1003641.

